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Assignment 1 C++

Section - A

Q.1 What is the purpose of # include <iostream> ?

Ans # include <io stream> is used to include the input-output stream library in C++. It allows the use of cin for input and cout for output.

Q.2 What is an Object and Class? Give one example.

Ans A class is a blueprint or template used to create objects. An object is an instance of a class.

E.g: If car is a class, then BMW is an object of class car.

Q.3 Define Abstraction in C++.

Ans Abstraction is the process of hiding internal implementation details and showing only essential features to the user.
It is achieved using classes and abstract classes in C++.

Q.4 Write the use of scope resolution operator and extraction operator.

Ans
- Scope resolution operator (::) is used to access global variables or class members outside the class.
- Extraction operator (>>) is used to take input from the user using cin.

Q.5 Write a C++ code to check whether a number is even or odd.

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Q.2

Ans

```
#include <iostream>
using namespace std;
int main (){
    int n;
    cin >> n;
    if (n%2 == 0)
        cout << "Even";
    else
        cout << "Odd";
    return 0;
}
```

Section - B

Q.1 Differentiate between Procedure Oriented Programming (POP) and Object Oriented Programming (OOP).

Ans

★ Procedure Oriented Programming	★ Object Oriented Programming
• Focuses on functions	• Focuses on object
• Data is less secure	• Data is more secure
• Uses top-down approch	• Uses bottom-up approch
• No concept of classes	• Uses classes and objects
• E.g: C	• Eg: C++

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Q.2 What do you mean by Abstract Data Type? Expalain with example.

Ans An Abstract Data Type (ADT) is a data type that defines the behaviour of data but hides its implementation. It specifies what operations can be performed, not how they are performed.

Example : A stack is a ADT where operations like push () and pop() are defined, but internal working is hidden.

Q.3 Write a program in C++ to implement arithmetic operators.

Ans

```
# include <io stream>
using namespace std;
int main (){
    int a=10, b=5;
    cout << "Addition :" << a+b << endl;
    cout << "Subtraction :" << a-b << endl;
    cout << "Multiplication:" << a*b << endl;
    cout << "Division :" << a/b << endl;
    cout << "Modulus :" << a%b << endl;
    return 0;
}
```

Section - C

Q.1 What do you mean by actual and formal parameters? Write a program to pass parameters to a function.

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Ans Actual Parameters are the values or variables passed to a function at the time of function call. They supply the data to the function.

Formal Parameters are the variables defined in the function definition that receive the values of actual parameters. They act as placeholders inside the function.

E.g : In sum(5, 10), 5 and 10 are actual parameters, while int a, int b in sum (int a, int b) are formal parameters.

```
#include <iostream>
using namespace std;

void add (int x, int y) { // formal parameters
    cout << "Sum = " << x+y;
}

int main () {
    add (5, 10); // actual parameters
    return 0;
}
```

Q.2 Write the characteristics of Object Oriented Programming. Explain Encapsulation with example.

Ans Characteristics of OOP :

- Class and Object
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism.
- Data Security

Encapsulation : It means binding data and functions together into a single unit called class. It protects data from unauthorized access.

Example :

```
class Student {
private :
    int marks;

public :
    void setMarks (int m) {
        marks = m;
    }
};
```

Q.3 Write a short note on :

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(a) Class and STL (Standard Template Library)

Ans A class is a user-defined data type that contains members and member function. STL is a collection of predefined classes and function such as vectors, lists, and algorithms that make programming easier.

(b) Constants, Variables and Keywords.

- Constants are fixed values that do not chang during program execution.
- Variables are used to store data values.
- Keywords are reserved words in C++ like int, return, if, while.

(c) Data Types in C++

Ans These specify the type of data that a variable can store. They help the compiler allocate memory and perform operations correctly.

Main Data types in C++ are:

1. Basic (Primitive) Data Types: These store simple values.
   E.g: int, float, char, double, bool.

2. Derived Data Types: These are derived from basic data type.
   E.g: Arrays, pointers, reference and functions.

3. User-defined Data Types: Thes are created by user.
   E.g: strut, union, class, enum.

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